Binomial Theorem

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Based on the lecture note uploaded on SPeCTRUM

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Learning Outcome

$(x+y)(x+y)(x+y)(x+y)$

At the end of this lecture, you should be able to:

$(x+y)^8$

□ Apply the binomial theorem to expand the binomial expression.

□ Apply the binomial theorem to determine the coefficient of a specific term in binomial expansion.

$(x+y)^8 = \_\,x^8 + \_\,x^7y + \_ + \cdots$

$+ \_\,y^8$

A binomial expression is the sum of two terms, such as $x + y$. The Binomial Theorem gives the coefficients of the expansion of powers of binomial expression $(x + y)^n$

The Binomial Theorem: Let x and y be variables, and let n be a positive integer. Then

general form.

$$(x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^r y^{n-r} = \overset{r=0}{\binom{n}{0}} y^n + \overset{r=1}{\binom{n}{1}} x y^{n-1} + \cdots + \binom{n}{r} x^r y^{n-r} + \cdots + \overset{r=n}{\binom{n}{n}} x^n.$$

Here, $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ (also denoted as ${}^nC_r$) is called a binomial coefficient.

Binomial coefficient

Find the value of the following binomial coefficients

1. $^5C_3 = \frac{5!}{3!(5-3)!} = \frac{5\times\cancel{4}^2\times\cancel{3}\times\cancel{2}\times\cancel{1}}{[\cancel{3}\times\cancel{2}\times\cancel{1}][\cancel{2}\times1]} = 10$

2. $^5C_2 = \frac{5!}{2!(5-2)!} = \frac{5\times\cancel{4}^2\times\cancel{3}\times\cancel{2}\times\cancel{1}}{[\cancel{2}\times1][\cancel{3}\times\cancel{2}\times\cancel{1}]} = 10$

$r = n-(n-r)$

$$^nC_r = \frac{n!}{r!(n-r)!} = \frac{n!}{(n-r)![n-(n-r)]!} = {}^nC_{n-r}$$

So, Binomial Theorem can also be written as

$$(x+y)^n = \sum_{r=0}^{n}\binom{n}{n-r}x^r y^{n-r} = \sum_{r=0}^{n}\binom{n}{r}x^r y^{n-r}$$

Pascal's Identity: Let n and r be positive integers with $n \geq r$. Then

$$\binom{n+1}{r} = \binom{n}{r-1} + \binom{n}{r}.$$

You will learn to prove this in SIM2013.

Pascal's Identity is the basis for a geometric arrangement of the binomial coefficients in a triangle called Pascal's triangle.

							1						
$n=1$						1		1					
$n=2$					1		2		1				
$n=3$				1		3		3		1			
$n=4$			1		4		6		4		1		
$n=5$		1		5		10		10		5		1	
$n=6$	1		6		15		20		15		6		1
$n=7$	1	7		21		35		35		21		7	1

$(x+y)^1 = 1x + 1y$

$(x+y)^2 = 1x^2 + 2xy + 1y^2$

$(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$

$(x+y)^5 = 1x^5y^0 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + 1x^0y^5$

$$(\underline{2x}+y)^4 = 1(2x)^4y^0 + 4(2x)^3y^1 + 6(2x)^2y^2 + 4(2x)^1y^3 + 1(2x)^0y^4$$

$$(2x-y)^3$$

$$\Downarrow$$

$$[(2x+\underline{(-y)})]^3 = 1(2x)^3(-y)^0 + 3(2x)^2(-y)^1 + 3(2x)^1(-y)^2 + 1(2x)^0(-y)^3$$

$$= 8x^3 - 3(4x^2)y + 3(2x)y^2 - y^3$$

Example

Expand the following

1. $(x-3)^5 = \binom{5}{0}x^5(-3)^0 + \binom{5}{1}x^4(-3)^1 + \overset{10}{\binom{5}{2}}x^3\overset{9}{(-3)^2} + \binom{5}{3}x^2(-3)^3$
   $+\binom{5}{4}x^1(-3)^4 + \binom{5}{5}x^0(-3)^5 = x^5 - 15x^4 + 90x^3$
   $-270x^2 + 405x - 243.$

2. $(x-2y)^4 = \binom{4}{0}x^4(-2y)^0 + \binom{4}{1}x^3(-2y)^1 + \binom{4}{2}x^2(-2y)^2$
   $+\binom{4}{3}x(-2y)^3 + \binom{4}{4}x^0(-2y)^4$
   $= x^4 - 8x^3y + 24x^2y^2 - 32xy^3 + 16y^4.$

Find the coefficient of $x^{\overset{r=5}{5}}y^2$ in the expansion of $(x+y)^{\overset{n=7}{7}}$. $= \binom{7}{5} = 21.$

Find the coefficient of $a^{\overset{r=5}{5}}b^2$ in the expansion of $(2a-3b)^{\overset{n=7}{7}}$.

In the expansion, we have

$\ldots + \binom{7}{5}(2a)^5(-3b)^2 + \ldots$

$\binom{7}{5}(2)^5(-3)^2 = 6048.$

Extra: $(x^2+y)^{3}$ n To find the coefficient of x^2

$$\binom{3}{0}(x^2)^3y^0+\binom{3}{1}(x^2)^2y^1+\binom{3}{2}(x^2)^1y^2+\binom{3}{3}(x^2)^0y^3$$

$r=2$

$x^4 \leftarrow r=1$